

# MATHEMATICAL METHODS II (MTH 202)

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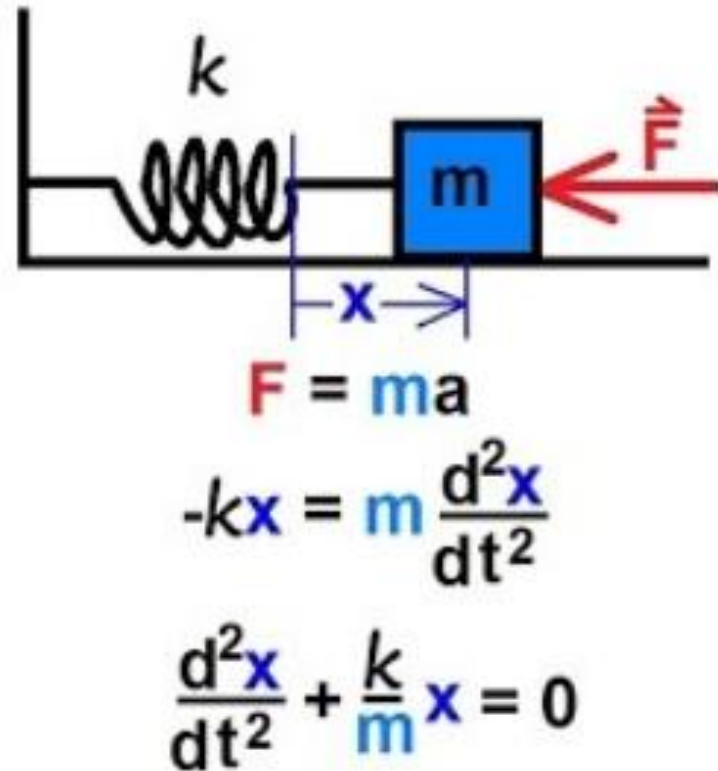
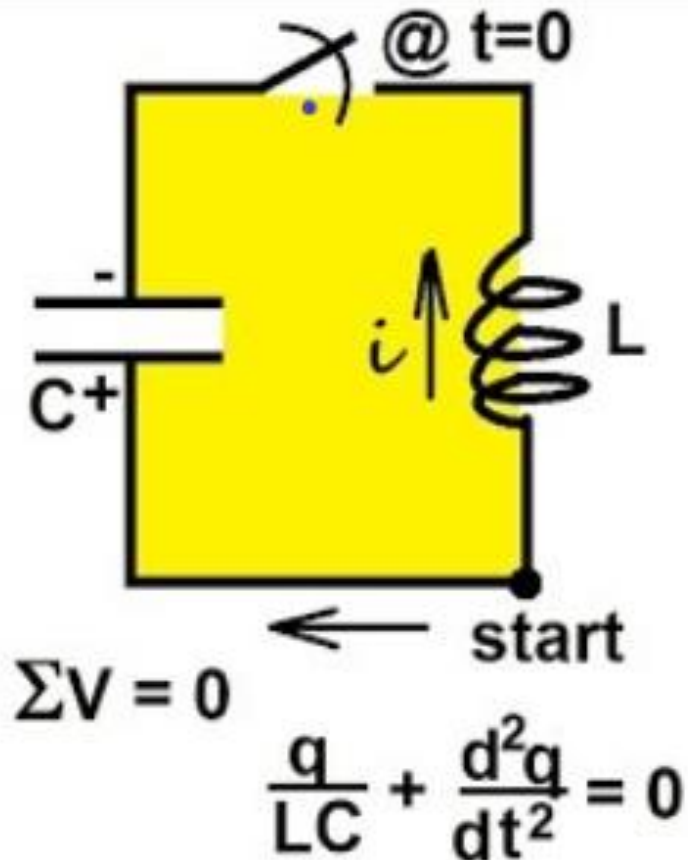


**SCHOOL OF  
SCIENCE AND  
TECHNOLOGY**

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**PAN-ATLANTIC UNIVERSITY**

# Application of Second Order ODEs



# General Linear Ordinary Differential Equation

The general linear differential equation is given by:

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_{n-1}(x) \frac{dy}{dx} + a_n(x)y = f(x)$$

- Where  $a_0(x)$ ,  $a_1(x)$ ,  $\cdots$ ,  $a_n(x)$  and  $f(x)$  are given functions of  $x$  or sometimes constants.
- Where  $f(x) = 0$ , It is said to be **Homogeneous**
- Where  $f(x) \neq 0$ , It is said to be **Inhomogeneous**



# Second Order ODE

The general linear second order ODE is given by:

$$a_0(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_2(x)y = f(x)$$

- If  $a_0(x)$ ,  $a_1(x)$ ,  $a_2(x)$  are constants, it is called a second order ODE with constant coefficients.
  - Where  $f(x) = 0$ , It is called a **Homogeneous Second order ODE with constant coefficients**
  - Where  $f(x) \neq 0$ , It is called a **Inhomogeneous Second order ODE with constant coefficients**



# Homogeneous Second Order ODE with Constant Coefficients

The homogeneous second order ODE with constant coefficients is given by:

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0$$

The **complementary solution** ( $y_c$ ) is obtained by using the trial solution

$$y = e^{mx}$$

Allowing the trial solution satisfy the second order differential equation we have the following:

$$\frac{dy}{dx} = me^{mx}$$

$$\frac{d^2 y}{dx^2} = m^2 e^{mx}$$



# Auxillary / Characteristic Equation

Substituting in the second order ODE, we obtain

$$a_0 m^2 e^{mx} + a_1 m e^{mx} + a_2 e^{mx} = 0$$

$$e^{mx} (a_0 m^2 + a_1 m + a_2) = 0$$

Implies the following:

$$e^{mx} \neq 0 \qquad a_0 m^2 + a_1 m + a_2 = 0$$

- The resulting quadratic equation is known as the **Characteristic equation**.
- The form of the solution is dependent on the nature of solution of the characteristic equation.

# Solution of Quadratic Equations

Given the quadratic equation

$$ax^2 + bx + c = 0$$

$$\Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The following are the types of roots and the condition where it occurs:

- Real and distinct Roots:  $b^2 - 4ac > 0$
- Equal Roots:  $b^2 - 4ac = 0$
- Complex Roots:  $b^2 - 4ac < 0$



# Solution of Second Order ODEs

Given the Characteristic equations

$$a_0 m^2 + a_1 m + a_2 = 0$$

- Obtain the roots and write the solution in the form:
- Real and distinct Roots ( $m_1 \neq m_2$ )

$$y = Ae^{m_1 x} + Be^{m_2 x}$$

- Equal Roots ( $m_1 = m_2 = m$ )

$$y = (A + Bx)e^{mx}$$

- Complex Roots ( $m = \alpha \pm i\beta$ )

$$y = e^{\alpha x}(A \cos \beta x + B \sin \beta x)$$



Example 1: Solve the ODE  $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$

$$y = e^{mx} \Rightarrow \frac{dy}{dx} = me^{mx} \Rightarrow \frac{d^2y}{dx^2} = m^2e^{mx}$$

Substitute in the ODE to obtain:

$$m^2e^{mx} + 3me^{mx} + 2e^{mx} = 0$$

$$e^{mx}(m^2 + 3m + 2) = 0$$

We have that  $e^{mx} \neq 0$ , therefore we obtain the characteristic equation

$$m^2 + 3m + 2 = 0$$



# Solve by Factorization method

$$m^2 + 3m + 2 = 0$$

$$(m + 1)(m + 2) = 0$$

$$\Rightarrow m = -1 \text{ or } -2$$

The roots are real and distinct, hence the solution takes the form

$$y = Ae^{-x} + Be^{-2x}$$



Example 2: Solve the ODE  $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$

$$y = e^{mx} \Rightarrow \frac{dy}{dx} = me^{mx} \Rightarrow \frac{d^2y}{dx^2} = m^2e^{mx}$$

Substitute in the ODE to obtain:

$$m^2e^{mx} - 6me^{mx} + 9e^{mx} = 0$$

$$e^{mx}(m^2 - 6m + 9) = 0$$

We have that  $e^{mx} \neq 0$ , therefore we obtain the characteristic equation

$$m^2 - 6m + 9 = 0$$



# Solve by Factorization method

$$m^2 - 6m + 9 = 0$$

$$(m - 3)^2 = 0$$

$$\Rightarrow m = 3 \text{ twice}$$

The roots are equal, hence the solution takes the form

$$y = (A + Bx)e^{3x}$$



Example 3: Solve the ODE  $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 5y = 0$

$$y = e^{mx} \Rightarrow \frac{dy}{dx} = me^{mx} \Rightarrow \frac{d^2y}{dx^2} = m^2e^{mx}$$

Substitute in the ODE to obtain:

$$m^2e^{mx} + 4me^{mx} + 5e^{mx} = 0$$

$$e^{mx}(m^2 + 4m + 5) = 0$$

We have that  $e^{mx} \neq 0$ , therefore we obtain the characteristic equation

$$m^2 + 4m + 5 = 0$$



# Solve by Quadratic Formula

$$m = \frac{-4 \pm \sqrt{16 - 20}}{2} = \frac{-4}{2} \pm \frac{\sqrt{16 - 20}}{2}$$

$$m = -2 \pm \frac{\sqrt{-4}}{2} = -2 \pm i \frac{\sqrt{4}}{2}$$

$$m = -2 \pm i$$

The roots are complex, hence the solution takes the form

$$y = e^{-2x}(A \cos x + B \sin x)$$



# Solve the following Second Order ODEs

1.  $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$

2.  $\frac{d^2y}{dx^2} + 9y = 0$

3.  $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 25y = 0$

4.  $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 13y = 0$

5.  $4\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 37y = 0$



# Inhomogeneous Second Order ODE with Constant Coefficients

The homogeneous second order ODE with constant coefficients is given by:

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = f(x).$$

The general solution:

$y = \text{complementary solution} + \text{particular solution}$

- The **complementary solution** is obtained from the characteristics equation of homogeneous equation.
- The **particular solution** is determined by RHS  $f(x)$



# Particular Solution (Method of Undetermined Coefficients)

- We shall give the rules for obtaining the particular solution of the second order ODE using the method of undetermined coefficients.
- In the following slides, we give rules for the form of the particular solution for:
  - Exponential functions ( $ae^{bx}$ )
  - Trigonometric functions ( $a\sin bx$  &  $a\cos bx$ )
  - Polynomials ( $ax^b$ )

## Rule (Exponential): If $f(x) = ae^{bx}$

- If  $f(x) = ae^{bx}$ , and if the characteristic equation has  $m = b$  as a root that occurs  $k$  times, then the particular solution is of the form:

$$y_p = Ax^k e^{bx}$$

is used, where  $A$  is a constant to be determined different from  $A$  in the complementary solution.

## Rule (Trigonometry): If $f(x) = a \cos bx / a \sin bx$

- If  $f(x) = a \sin bx$  or  $a \cos bx$ , and if  $m^2 + b^2$  is a factor of the characteristic equation in which this factor may occur  $k$  times, then the particular solution is of the form:

$$y_p = x^k (A \sin bx + B \cos bx)$$

where constants  $a$  and  $b$  are to be determined.

## Rule (Polynomials): If $f(x) = ax^b$

- If  $f(x) = ax^b$ , where  $a$  and  $b$  are constants, and if the characteristic equation has  $m = 0$  as a root for which it occurs  $k$  times, then the particular solution is of the form:

$$y_p = x^k (A_b x^b + A_{b-1} x^{b-1} + \cdots + A_1 x + A_0)$$

where  $A_i, i = 1, 2, \dots, n$  are constants.

Example 1. Solve the ODE  $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = e^{2x}$

Characteristics equation:  $m^2 - 3m + 2 = 0$

$$(m - 1)(m - 2) = 0 \Rightarrow m = 1 \text{ or } 2$$

Complementary solution  $\Rightarrow y_c = Ae^x + Be^{2x}$

**Particular solution with**  $f(x) = 1e^{2x}$   
 $m = 2$  appear  $k = 1$  times

Therefore, the trial solution  $y_p$  is

$$y_p = Ax^1e^{2x} = Axe^{2x}$$

- We seek coefficient  $A$  such that  $y_p$  satisfy the ODE

$$y'_p = Ae^{2x} + 2Axe^{2x}$$

$$y''_p = 4Ae^{2x} + 4Axe^{2x}$$

The ODE becomes

$$4Ae^{2x} + 4Axe^{2x} - 3(Ae^{2x} + 2Axe^{2x}) + 2(Axe^{2x}) = e^{2x}$$

Which simplifies to

$$Ae^{2x} = e^{2x} \Rightarrow A = 1$$

Therefore

$$y_p = xe^{2x}$$

$$y = y_c + y_p$$

Hence,

$$y = Ae^x + Be^{2x} + xe^{2x}$$

Example 2. Solve the ODE  $\frac{d^2y}{dx^2} + 4y = 3 \sin 2x$

Characteristics equation:  $m^2 + 4 = 0 \Rightarrow m = \pm 2i$

Complex root  $(\alpha \pm i\beta) \Rightarrow e^{\alpha x}(A \cos \beta x + B \sin \beta x)$

Complementary solution  $\Rightarrow y_c = A \cos 2x + B \sin 2x$

**Particular solution with**  $f(x) = 3 \sin 2x$   
 $m^2 + 2^2$  appear  $k = 1$  times

Therefore, the trial solution  $y_p$  is

$$y_p = x^1(A \sin 2x + B \cos 2x) = x(A \sin 2x + B \cos 2x)$$

- We seek coefficient  $A$  &  $B$  such that  $y_p$  satisfy the ODE

$$y'_p = x(2A \cos 2x - 2B \sin 2x) + (A \sin 2x + B \cos 2x)$$

$$y''_p = -4Ax \sin 2x - 4Bx \cos 2x + 4A \cos 2x - 4B \sin 2x$$

The ODE simplifies to

$$4A \cos 2x - 4B \sin 2x = 3 \sin 2x$$

$$\Rightarrow 4A = 0 \qquad \Rightarrow -4B = 3$$

$$\Rightarrow A = 0 \qquad \Rightarrow B = -\frac{3}{4}$$

Therefore

$$y_p = x \left( 0 \sin 2x - \frac{3}{4} \cos 2x \right) = -\frac{3}{4} x \cos 2x$$

$$y = y_c + y_p$$

Hence,

$$y = A \cos 2x + B \sin 2x - \frac{3}{4} x \cos 2x$$

Example 3. Solve the ODE  $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 9x^2 + 1$

Characteristics equation:  $m^2 - 2m - 3 = 0$

$$(m + 1)(m - 3) = 0 \Rightarrow m = -1 \text{ or } 3$$

Complementary solution  $\Rightarrow y_c = Ae^{-x} + Be^{3x}$

**Particular solution with**  $f(x) = 9x^2 + 1$   
 $m = 0$  appear  $k = 0$  times

Therefore, the trial solution  $y_p$  is

$$y_p = x^0(A_2x^2 + A_1x + A_0) = A_2x^2 + A_1x + A_0$$

- We seek  $A_0$ ,  $A_1$ , &  $A_2$  such that  $y_p$  satisfy the ODE

$$y'_p = 2A_2x + A_1$$

$$y''_p = 2A_2$$

The ODE simplifies to

$$[-3A_0 - 2A_1 + 2A_2] - [3A_1 + 4A_0]x - [3A_2]x^2 = 1 + 9x^2$$

$$\Rightarrow \begin{cases} -3A_0 - 2A_1 + 2A_2 = 1 \\ -3A_1 - 4A_0 = 0 \\ -3A_2 = 9 \end{cases} \Rightarrow \begin{cases} A_0 = -5 \\ A_1 = 4 \\ A_2 = -3 \end{cases}$$

Therefore

$$y_p = -3x^2 + 4x - 5$$

$$y = y_c + y_p$$

Hence,

$$y = Ae^{-x} + Be^{3x} - 3x^2 + 4x - 5$$

# Assignment / Classwork

Solve the following ODE:

1.  $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^{4x}$

2.  $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 4x^2$

3.  $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 4y = \cos x$

4.  $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 10y = 5x + e^{-2x}$

5.  $\frac{d^2y}{dx^2} + \frac{dy}{dx} = 4\sin x + 2x^2$